SPHOORTHY ENGINEERING COLLEGE | *(UGC Autonomous — NAAC Grade A)*

Data Warehousing & Data Mining: ETL, OLAP & Pattern Discovery

Department: CSE DS	**Course / Subject:** Data Warehousing & Mining (DS501PC)
Semester: Semester 5	**Regulation:** Autonomous (SR23 / SR24)

Course Overview & Key Focus: Data warehouse architectures, Star & Snowflake schemas, OLAP operations (Roll-up, Drill-down, Slice-and-dice), Apriori association rule mining, and FP-Growth.

Unit 1: Data Warehousing Concepts

Operational databases vs Data Warehouses, Kimball vs Inmon architectures, ETL (Extract, Transform, Load) pipelines, and Data staging areas.

Unit 2: Multidimensional Data Modeling

Fact tables, Dimension tables, Star schema, Snowflake schema, Fact Constellation schema, Measures (Additive, Semi-additive, Non-additive).

Unit 3: OLAP Systems & Operations

MOLAP vs ROLAP vs HOLAP, OLAP operations: Roll-up, Drill-down, Slice, Dice, Pivot, and Data cube computation techniques.

Unit 4: Association Rule Mining

Market Basket Analysis, Support, Confidence, Lift metrics, Apriori algorithm candidate generation, Apriori property, and FP-Tree growth algorithm.

Unit 5: Advanced Classification & Clustering for Mining

Decision tree induction (ID3, C4.5, Gini index), Rule-based classification, Cluster evaluation (Silhouette score), and Mining text and graph streams.